E-Waste Battery Extraction Robot

Bill of Materials Assessment Report

1. Overview

This report presents a structured Bill of Materials (BoM) assessment for the E-Waste Battery Extraction Robot project 7. The robot platform under evaluation is the UFACTORY xARM 7 a 7-degree-of-freedom collaborative robotic arm currently operational in the robotics lab which suits are project needs for this trimester. The assessment covers component selection, pricing, and laboratory availability across two thermal management configurations.

The two configurations evaluated are:

1. **Configuration 1** Hot (Heating): Utilises an Electric Hot Plate as the thermal element for heating-based disassembly tasks with the mobile phones.
2. **Configuration 2** Cold (Cooling): Utilises a Liquid Cold Plate, Coolant (Syltherm XLT), and a Recirculating Chiller for temperature-controlled cooling operations to stiffen the adhesive on battery.

Components shared across both configurations include the xARM 7 robotic arm, vacuum gripper end-effector, 6-axis force torque sensor, Intel Realsense D435 depth camera, and a Habotest HT820 thermal monitoring camera.

2. Configuration 1 Hot (Heating)

This configuration is designed for thermal softening or heating-assisted disassembly processes. The Electric Hot Plate serves as the thermal element, providing a low-cost and straightforward heating solution. Most components in this configuration are already available at the Burwood Campus Robotics Lab, minimising procurement lead time.

Component	Item / Part	Price (AUD)	Available at Lab
Robotic Arm	UFACTORY xArm 7	AUD 15,403.89	**Yes**
End-Effector	UFACTORY xArm Vacuum Gripper	AUD 802.71	**Yes**
End-Effector Sensing	UFACTORY 6-Axis Force Torque Sensor	AUD 4,443.85	**Yes**
Thermal Element	Electric Hot Plate	AUD 81.99	**Procure**
Camera	Intel Realsense D435	AUD 822.46	**Yes**
Thermal Camera (Monitoring)	Habotest HT820	AUD 233.50	**Procure**

Note: Items marked ' Procure' are not currently available in the lab and will require purchase prior to deployment.

3. Configuration 2 Cold (Cooling)

The cooling configuration is designed for scenarios requiring temperature reduction during disassembly for example, cooling solder joints or thermally sensitive electronic components. This configuration involves a more complex and cost-intensive thermal subsystem comprising a Liquid Cold Plate, Syltherm XLT coolant, and a Recirculating Chiller. All three cooling-specific components require procurement.

Component	Item / Part	Price (AUD)	Available at Lab
Robotic Arm	UFACTORY xArm 7	AUD 15,403.89	Yes
End-Effector	UFACTORY xArm Vacuum Gripper	AUD 802.71	Yes
End-Effector Sensing	UFACTORY 6-Axis Force Torque Sensor	AUD 4,443.85	Yes
Thermal Element	Liquid Cold Plate	AUD 191.71	Procure
	Coolant (Syltherm XLT) – 5 gal	AUD 1,409.66	Procure
	Recirculating Chiller	AUD 8,917.44	Procure
Camera	Intel Realsense D435	AUD 822.46	Yes
Thermal Camera (Monitoring)	Habotest HT820	AUD 233.50	Procure

Note: The cooling subsystem represents the most significant additional cost in this configuration (~AUD 10,518.81). Items marked 'Yes' are already located at the Burwood Campus Robotics Lab.

4. Cost Summary

Configuration	Total Estimated Cost	Procurement Required
Configuration 1 – Hot (Heating)	AUD 21,788.40	AUD 315.49
Configuration 2 – Cold (Cooling)	AUD 32,025.51	AUD 10,752.31

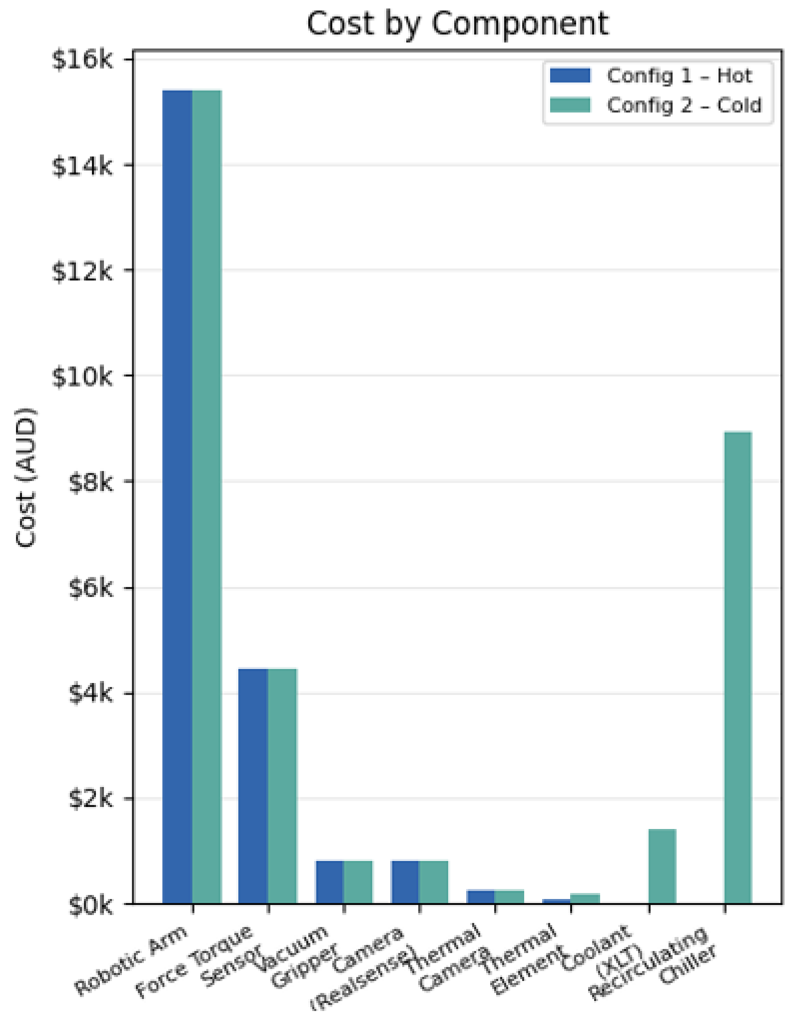

Figure 1: Graph made using google colab

5. Research Contributions

Researcher	Role / Contribution
Vinh	Research Contribution E-Waste Robot BoM Assessment
Tarunjeet	Research Contribution Cost Analysis Assessment